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- The DFT transforms an N -point sequence $x(n)$, $0 \leq n \leq N - 1$ into the N -point discrete Fourier domain samples $X(k)$, $0 \leq k \leq N - 1$.
- DFT is obtained by sampling the DTFT in the frequency domain (or the z -transform on the unit circle).

DFT contd...

- The N -point *discrete Fourier transform* (DFT) of a length- N signal is given by

$$X(k) \triangleq \text{DFT}[x(n)] = \sum_{n=0}^{N-1} x(n)e^{-j\frac{2\pi}{N}kn} \quad 0 \leq k \leq N-1$$

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DFT contd...

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- Defining a term

$$W_N = e^{-j \frac{2\pi}{N}} = \cos\left(\frac{2\pi}{N}\right) - j \sin\left(\frac{2\pi}{N}\right)$$

which is known as *twiddle factor*.

DFT contd...

- The formulas for DFT and IDFT may be expressed as (in terms of twiddle factor)

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn} \quad 0 \leq k \leq N-1$$

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-kn} \quad 0 \leq n \leq N-1$$

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DFT contd...

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- **Example:** Compute the 4-point DFT of $x(n)$.

$$x(n) = \{\underset{\uparrow}{1}, 1, 1, 1\} = \begin{cases} 1 & 0 \leq n \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

DFT Example contd...

● **Solution:** By definition, the 4-point DFT is given by

$$\begin{aligned} X(k) &= \sum_{n=0}^3 x(n)e^{-j\frac{2\pi}{4}kn} \quad 0 \leq k \leq 3 \\ &= x(0)e^{-j\frac{2\pi}{4}0k} + x(1)e^{-j\frac{2\pi}{4}k} + x(2)e^{-j\frac{2\pi}{4}2k} + x(3)e^{-j\frac{2\pi}{4}3k} \end{aligned}$$

$$X(k) = 1 + e^{-j\frac{\pi}{2}k} + e^{-j\pi k} + e^{-j\frac{3\pi}{2}k}$$

For $k = 0$: $X(0) = 1 + 1 + 1 + 1 = 4$

For $k = 1$: $X(1) = 1 + e^{-j\frac{\pi}{2}} + e^{-j\pi} + e^{-j\frac{3\pi}{2}} = 1 - j - 1 + j = 0$

For $k = 2$: $X(2) = 1 + e^{-j\pi} + e^{-j2\pi} + e^{-j3\pi} = 1 - 1 + 1 - 1 = 0$

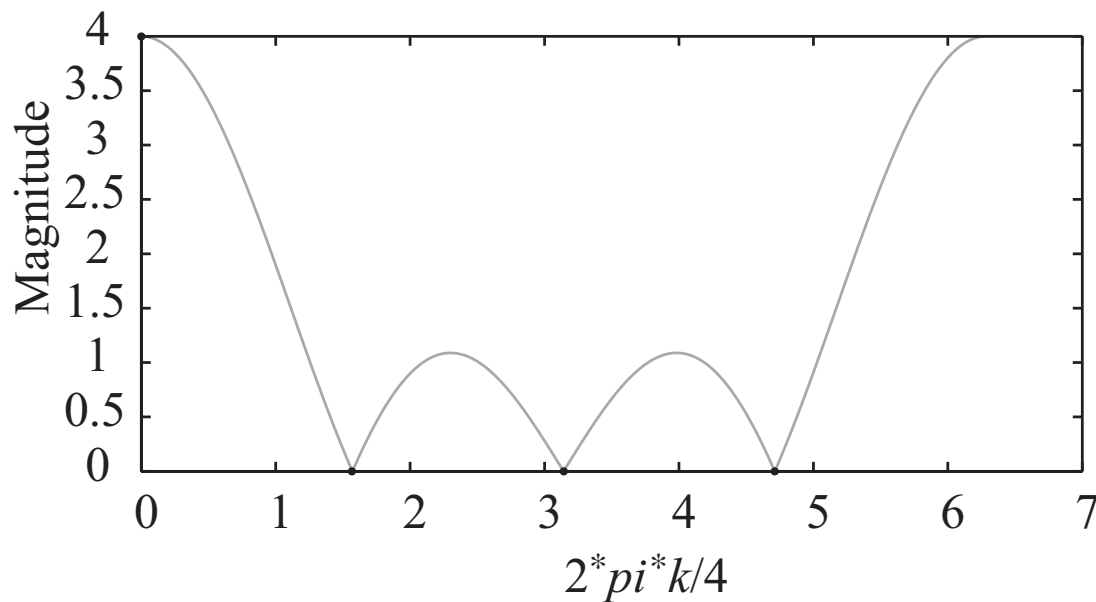
For $k = 3$: $X(3) = 1 + e^{-j\frac{3\pi}{2}} + e^{-j3\pi} + e^{-j\frac{9\pi}{2}} = 1 + j - 1 - j = 0$

Therefore,

$$X(k) = \{\underset{\uparrow}{4}, 0, 0, 0\}$$

DFT Example contd...

- The plot of DFT values is shown in Figure. The plot of $|X(e^{j\omega})|$ is also shown for comparison. From the plot of Figure, we observe that $X(k)$ correctly gives 4-samples of $X(e^{j\omega})$, but it has one nonzero sample. By looking at the 4-point $x(n)$, which contains all 1's, one must conclude that its periodic extension is $x_p(n) = 1$ for all n , which is a constant (or a DC) signal. This is what is predicted by the DFT $X(k)$, which has a nonzero sample at $k = 0$ (or $\omega = 0$) and has no values at other frequencies.



Zero Padding

- **Zero Padding:** Zero-padding refers to the operation of extending a signal of length- N to a length $M > N$ by appending $M - N$ zero samples to the given signal.
- The length of the signal can be increased from N to M by padding $M - N$ zeros as follows.

$$x_1(n) = \begin{cases} x(n) & 0 \leq n \leq N - 1 \\ 0 & N \leq n \leq M - 1 \end{cases} \quad (1)$$

Here $x_1(n)$ denotes the zero-padded version of $x(n)$.

- The effect of zero padding is to interpolate between the original points of the spectrum. The longer DFT provides closely spaced samples of the DTFT of the original signal.

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$$x_1(n) = \begin{cases} x(n) & 0 \leq n \leq N - 1 \\ 0 & N \leq n \leq M - 1 \end{cases} \quad (2)$$

Here $x_1(n)$ denotes the zero-padded version of $x(n)$.

- The effect of zero padding is to interpolate between the original points of the spectrum. The longer DFT provides closely spaced samples of the DTFT of the original signal.
- **Example:** Compute 8-point DFT of the sequence $x_1(n) = \{1, 1, 1, 1, 0, 0, 0, 0\}$.

DFT Example contd...

● **Solution:** Given that $N = 8$. By definition, the N -point DFT is given by

$$X_1(k) = \sum_{n=0}^{N-1} x(n) e^{-j \frac{2\pi}{N} kn} \quad 0 \leq k \leq N-1$$

$$X_1(k) = \sum_{n=0}^7 x_1(n) e^{-j \frac{2\pi}{8} kn} \quad 0 \leq k \leq 7$$

$$\begin{aligned} &= x(0)e^{-j \frac{2\pi}{8} 0k} + x(1)e^{-j \frac{2\pi}{8} k} + x(2)e^{-j \frac{2\pi}{8} 2k} + x(3)e^{-j \frac{2\pi}{8} 3k} \\ &\quad + x(4)e^{-j \frac{2\pi}{8} 4k} + x(5)e^{-j \frac{2\pi}{8} 5k} + x(6)e^{-j \frac{2\pi}{8} 6k} + x(7)e^{-j \frac{2\pi}{8} 7k} \\ &= 1 + e^{-j \frac{\pi}{4} k} + e^{-j \frac{\pi}{2} k} + e^{-j \frac{3\pi}{4} k} + 0 + 0 + 0 + 0 \end{aligned}$$

$$X_1(k) = 1 + e^{-j \frac{\pi}{4} k} + e^{-j \frac{\pi}{2} k} + e^{-j \frac{3\pi}{4} k}$$

For $k = 0$: $X_1(0) = 4$

For $k = 1$: $X_1(1) = 1 + e^{-j \frac{\pi}{4}} + e^{-j \frac{\pi}{2}} + e^{-j \frac{3\pi}{4}} = 1 - j2.414 = 2.613e^{-j67.5^\circ}$

For $k = 2$: $X_1(2) = 1 + e^{-j \frac{\pi}{2}} + e^{-j \pi} + e^{-j \frac{3\pi}{2}} = 1 - j - 1 + j = 0$

DFT Example contd...

$$\text{For } k = 3: \quad X_1(3) = 1 + e^{-j\frac{3\pi}{4}} + e^{-j\frac{3\pi}{2}} + e^{-j\frac{9\pi}{4}} = 1 - j0.414 = 1.082e^{-j22.5^\circ}$$

$$\text{For } k = 4: \quad X_1(4) = 1 + e^{-j\pi} + e^{-j2\pi} + e^{-j3\pi} = 1 - 1 + 1 - 1 = 0$$

$$\text{For } k = 5: \quad X_1(5) = 1 + e^{-j\frac{5\pi}{4}} + e^{-j\frac{5\pi}{2}} + e^{-j\frac{15\pi}{4}} = 1 + j0.414 = 1.082e^{j22.5^\circ}$$

$$\text{For } k = 6: \quad X_1(6) = 1 + e^{-j\frac{3\pi}{2}} + e^{-j3\pi} + e^{-j\frac{9\pi}{2}} = 1 + j - 1 - j = 0$$

$$\text{For } k = 7: \quad X_1(7) = 1 + e^{-j\frac{7\pi}{4}} + e^{-j\frac{7\pi}{2}} + e^{-j\frac{21\pi}{4}} = 1 + j2.414 = 2.613e^{j67.5^\circ}$$

Hence

$$|X_1(k)| = \{4, 2.613, 0, 1.082, 0, 1.082, 0, 2.613\}$$

$$\angle X_1(k) = \{0, -67.5, 0, -22.5, 0, 22.5, 0, 67.5\}$$

Note that the sequence $x_1(n)$ is zero-padded version of $x(n)$ given in the earlier Example.

DFT as a Linear Transformation (Matrix Formulation)

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn} \quad 0 \leq k \leq N-1$$

N -point DFT can be expressed in matrix form as

$$\underbrace{\begin{bmatrix} X(0) \\ X(1) \\ \vdots \\ X(N-1) \end{bmatrix}}_{\mathbf{X}} = \underbrace{\begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & W_N^1 & W_N^2 & \cdots & W_N^{N-1} \\ 1 & W_N^2 & W_N^4 & \cdots & W_N^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & W_N^{N-1} & W_N^{2(N-1)} & \cdots & W_N^{(N-1)(N-1)} \end{bmatrix}}_{\mathbf{W}_N} \underbrace{\begin{bmatrix} x(0) \\ x(1) \\ \vdots \\ x(N-1) \end{bmatrix}}_{\mathbf{x}}$$

$$\mathbf{X} = \mathbf{W}_N \mathbf{x}$$

DFT as a Linear Transformation (Matrix Formulation) contd...

- The IDFT can be expressed in matrix form as

$$\mathbf{x} = \mathbf{W}_N^{-1} \mathbf{X}$$

where $\mathbf{W}_N^{-1}$ is the $N \times N$ IDFT matrix given by

$$\mathbf{W}_N^{-1} = \frac{1}{N} \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & W_N^{-1} & W_N^{-2} & \dots & W_N^{-(N-1)} \\ 1 & W_N^{-2} & W_N^{-4} & \dots & W_N^{-2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & W_N^{-(N-1)} & W_N^{-2(N-1)} & \dots & W_N^{-(N-1)(N-1)} \end{bmatrix}$$

Properties of the DFT

● **Periodicity:** If $x(n)$ and $X(k)$ are N -point DFT pair, then

$$X(k + N) = X(k) \quad \text{for all } k \quad \text{and} \quad x(n + N) = x(n) \quad \text{for all } n$$

● **Proof:** By definition of DFT

$$\begin{aligned} X(k) &= \sum_{n=0}^{N-1} x(n) W_N^{kn} \\ X(k + N) &= \sum_{n=0}^{N-1} x(n) W_N^{(k+N)n} = \sum_{n=0}^{N-1} x(n) W_N^{kn} \underbrace{W_N^{Nn}}_{=1} = \sum_{n=0}^{N-1} x(n) W_N^{kn} = X(k) \end{aligned}$$

By definition of IDFT

$$\begin{aligned} x(n) &= \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-kn} \\ x(n + N) &= \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-k(n+N)} = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-kn} \underbrace{W_N^{-kN}}_{=1} = x(n) \end{aligned}$$

Properties of the DFT contd...

● **Linearity:** If

$$x_1(n) \xleftrightarrow[N]{\text{DFT}} X_1(k) \quad \text{and} \quad x_2(n) \xleftrightarrow[N]{\text{DFT}} X_2(k)$$

then

$$ax_1(n) + bx_2(n) \xleftrightarrow[N]{\text{DFT}} aX_1(k) + bX_2(k)$$

● **Proof:** By definition

$$\begin{aligned} DFT[ax_1(n) + bx_2(n)] &= \sum_{n=0}^{N-1} [ax_1(n) + bx_2(n)] W_N^{kn} \\ &= a \underbrace{\sum_{n=0}^{N-1} x_1(n) W_N^{kn}}_{X_1(k)} + b \underbrace{\sum_{n=0}^{N-1} x_2(n) W_N^{kn}}_{X_2(k)} = aX_1(k) + bX_2(k) \end{aligned}$$

Properties of the DFT contd...

● **Circular Time Shifting:** In general the circular shifted version of $x(n)$ is given by

$$x(n) = [x(0), x(1), x(2), \dots, x(N-2), x(N-1)]$$

$$x(\langle n-1 \rangle_N) = [x(N-1), x(0), x(1), \dots, x(N-3), x(N-2)]$$

$$x(\langle n-2 \rangle_N) = [x(N-2), x(N-1), x(0), \dots, x(N-4), x(N-3)]$$

$$x(\langle n-n_0 \rangle_N) = [x(N-n_0), x(N-n_0+1), \dots, x(N-n_0-1)]$$

$$x(\langle n-N \rangle_N) = [x(0), x(1), x(2), \dots, x(N-2), x(N-1)]$$

For example, let $x(n) = \{\underset{\uparrow}{1}, 2, 3, 4, 5\}$, $N = 5$ and $n_0 = 2$. The circular shifted signal $x_c(n) = x(\langle n-2 \rangle_5) = \{\underset{\uparrow}{4}, 5, 1, 2, 3\}$ is shown in Figure.

Properties of the DFT contd...

$$x_c(n) = x(\langle n - 2 \rangle_5) \quad 0 \leq n \leq 4$$

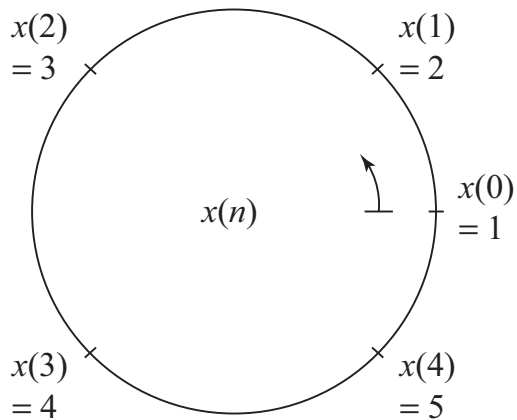
$$x_c(0) = x(\langle -2 \rangle_5) = x(-2 + 5) = x(3) = 4$$

$$x_c(1) = x(\langle -1 \rangle_5) = x(-1 + 5) = x(4) = 5$$

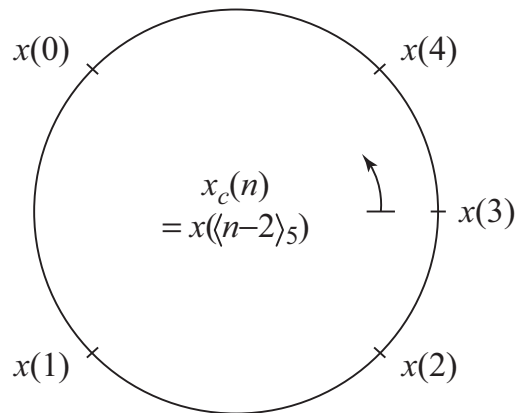
$$x_c(2) = x(\langle 0 \rangle_5) = x(0) = 1$$

$$x_c(3) = x(\langle 1 \rangle_5) = x(1) = 2$$

$$x_c(4) = x(\langle 2 \rangle_5) = x(2) = 3$$



(a)



(b)

Properties of the DFT contd...

● If

$$x(n) \xleftrightarrow[N]{\text{DFT}} X(k)$$

then

$$x(\langle n - n_o \rangle_N) \xleftrightarrow[N]{\text{DFT}} X(k) e^{-j \frac{2\pi}{N} k n_o} = X(k) W_N^{k n_o}$$

and

$$x(\langle n + n_o \rangle_N) \xleftrightarrow[N]{\text{DFT}} X(k) e^{j \frac{2\pi}{N} k n_o} = X(k) W_N^{-k n_o}$$

● **Proof:** By definition

$$\text{DFT}[x(\langle n - n_o \rangle_N)] = \sum_{n=0}^{N-1} x(\langle n - n_o \rangle_N) e^{-j \frac{2\pi}{N} k n}$$

Properties of the DFT contd...

we can define

$$x(\langle n - n_o \rangle_N) = \begin{cases} x(n - n_o) & n_o \leq n \leq N - 1 \\ x(n - n_o + N) & 0 \leq n \leq n_o - 1 \end{cases}$$

and therefore

$$\begin{aligned} \text{DFT}[x(\langle n - n_o \rangle_N)] &= \sum_{n=0}^{n_o-1} x(n - n_o + N) e^{-j \frac{2\pi}{N} kn} + \sum_{n=n_o}^{N-1} x(n - n_o) e^{-j \frac{2\pi}{N} kn} \\ \text{DFT}[x(\langle n - n_o \rangle_N)] &= \sum_{r=N-n_o}^{N-1} x(r) e^{-j \frac{2\pi}{N} k(r+n_o)} + \sum_{r=0}^{N-n_o-1} x(r) e^{-j \frac{2\pi}{N} k(r+n_o)} \\ &= \sum_{r=0}^{N-1} x(r) e^{-j \frac{2\pi}{N} k(r+n_o)} = \left(\sum_{r=0}^{N-1} x(r) e^{-j \frac{2\pi}{N} kr} \right) e^{-j \frac{2\pi}{N} kn_o} \\ &= X(k) e^{-j \frac{2\pi}{N} kn_o} = X(k) W_N^{kn_o} \end{aligned}$$

Properties of the DFT contd...

● Circular Frequency Shifting: If

$$x(n) \xleftrightarrow[N]{\text{DFT}} X(k)$$

then

$$e^{j\frac{2\pi}{N}k_0n}x(n) = W_N^{-k_0n}x(n) \xleftrightarrow[N]{\text{DFT}} X(\langle k - k_0 \rangle_N)$$

and

$$e^{-j\frac{2\pi}{N}k_0n}x(n) = W_N^{k_0n}x(n) \xleftrightarrow[N]{\text{DFT}} X(\langle k + k_0 \rangle_N)$$

● **Proof:** By definition

$$\text{IDFT}[X(\langle k - k_0 \rangle_N)] = \frac{1}{N} \sum_{k=0}^{N-1} X(\langle k - k_0 \rangle_N) e^{j\frac{2\pi}{N}kn}$$

Properties of the DFT contd...

Since

$$X(\langle k - k_o \rangle_N) = \begin{cases} X(k - k_o) & k_o \leq k \leq N - 1 \\ X(k - k_o + N) & 0 \leq k \leq k_o - 1 \end{cases}$$

we have

$$\begin{aligned} \text{IDFT}[X(\langle k - k_o \rangle_N)] &= \frac{1}{N} \sum_{k=0}^{k_o-1} X(k - k_o + N) e^{j \frac{2\pi}{N} kn} + \frac{1}{N} \sum_{k=k_o}^{N-1} X(k - k_o) e^{j \frac{2\pi}{N} kn} \\ \text{IDFT}[X(\langle k - k_o \rangle_N)] &= \frac{1}{N} \sum_{r=N-k_o}^{N-1} X(r) e^{j \frac{2\pi}{N} n(r+k_o)} + \frac{1}{N} \sum_{r=0}^{N-k_o-1} X(r) e^{j \frac{2\pi}{N} n(r+k_o)} \\ &= \frac{1}{N} \sum_{r=0}^{N-1} X(r) e^{j \frac{2\pi}{N} n(r+k_o)} = \left(\frac{1}{N} \sum_{r=0}^{N-1} X(r) e^{j \frac{2\pi}{N} nr} \right) e^{j \frac{2\pi}{N} nk_o} \\ &= x(n) e^{j \frac{2\pi}{N} nk_o} = x(n) W_N^{-nk_o} \end{aligned}$$

Properties of the DFT contd...

● Circular Time Reversal: If

$$x(n) \xleftrightarrow[N]{\text{DFT}} X(k)$$

$$\text{then } x(\langle -n \rangle_N) \xleftrightarrow[N]{\text{DFT}} X(\langle -k \rangle_N)$$

$$x(N - n) \xleftrightarrow[N]{\text{DFT}} X(N - k)$$

● Proof: By definition

$$\begin{aligned} \text{DFT}[x(\langle -n \rangle_N)] &= \sum_{n=0}^{N-1} x(\langle -n \rangle_N) e^{-j \frac{2\pi}{N} kn} \\ &= x(0) + \sum_{r=1}^{N-1} x(r) e^{j \frac{2\pi}{N} kr} = \sum_{r=0}^{N-1} x(r) e^{j \frac{2\pi}{N} kr} \\ &= \sum_{r=0}^{N-1} x(r) e^{-j \frac{2\pi}{N} (-k)r} = X(\langle -k \rangle_N) = X(N - k) \end{aligned}$$

Properties of the DFT contd...

● Conjugation and Conjugate Symmetry (Symmetry Properties): If

$$x(n) \xleftrightarrow[N]{\text{DFT}} X(k)$$

$$\text{then } x^*(n) \xleftrightarrow[N]{\text{DFT}} X^*(\langle -k \rangle_N) = X^*(N - k)$$

$$\text{and } x^*(N - n) = x^*(\langle -n \rangle_N) \xleftrightarrow[N]{\text{DFT}} X^*(k)$$

● **Proof:** By definition

$$\begin{aligned} \text{DFT}[x^*(n)] &= \sum_{n=0}^{N-1} x^*(n) e^{-j \frac{2\pi}{N} kn} \\ &= \left[\sum_{n=0}^{N-1} x(n) e^{j \frac{2\pi}{N} kn} \right]^* = \left[\sum_{n=0}^{N-1} x(n) e^{-j \frac{2\pi}{N} (-k)n} \right]^* \\ &= [X(\langle -k \rangle_N)]^* = X^*(\langle -k \rangle_N) = X^*(N - k) \end{aligned}$$

Properties of the DFT contd...

Similarly, by definition

$$\begin{aligned}\text{IDFT}[X^*(k)] &= \frac{1}{N} \sum_{k=0}^{N-1} X^*(k) e^{j \frac{2\pi}{N} kn} = \left[\frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{-j \frac{2\pi}{N} kn} \right]^* = \left[\frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j \frac{2\pi}{N} k(-n)} \right]^* \\ &= [x(\langle -n \rangle_N)]^* = x^*(\langle -n \rangle_N) = x^*(N - n)\end{aligned}$$

Case I: If $x(n)$ is real, that is, if

$$x(n) = x^*(n)$$

then

$$\text{DFT}[x(n)] = \text{DFT}[x^*(n)]$$

$$X(k) = X^*(\langle -k \rangle_N) = X^*(N - k)$$

$$X^*(k) = X(\langle -k \rangle_N) = X(N - k) \quad (\text{conjugate symmetric})$$

The DFT of a real signal is *conjugate symmetric*.

Properties of the DFT contd...

🟢 **Case II:** If $x(n)$ is *real* & *circularly even*, that is, if

$$x(n) = x^*(n) = x(\langle -n \rangle_N)$$

then

$$\text{DFT}[x(n)] = \text{DFT}[x^*(n)] = \text{DFT}[x(\langle -n \rangle_N)]$$

$$X(k) = X^*(\langle -k \rangle_N) = X(\langle -k \rangle_N) \quad (\text{real \& circularly even})$$

$$X(k) = X^*(N - k) = X(N - k)$$

If $x(n)$ is real & circularly even then its DFT $X(k)$ is also real & circularly even.

Properties of the DFT contd...

🟢 **Case III:** If $x(n)$ is *real & circularly odd*, that is, if

$$x(n) = x^*(n) = -x(\langle -n \rangle_N)$$

then

$$\text{DFT}[x(n)] = \text{DFT}[x^*(n)] = \text{DFT}[-x(\langle -n \rangle_N)]$$

$$X(k) = X^*(\langle -k \rangle_N) = -X(\langle -k \rangle_N) \quad (\text{imaginary \& circularly odd})$$

$$X(k) = X^*(N - k) = -X(N - k)$$

If $x(n)$ is real & circularly odd then its DFT $X(k)$ is imaginary & circularly odd.

Properties of the DFT contd...

🟢 **Case IV:** If $x(n)$ is imaginary, that is, if

$$x(n) = -x^*(n)$$

then

$$\text{DFT}[x(n)] = \text{DFT}[-x^*(n)]$$

$$X(k) = -X^*(\langle -k \rangle_N) = -X^*(N - k)$$

$$X^*(k) = -X(\langle -k \rangle_N) = -X(N - k) \quad (\text{conjugate antisymmetric})$$

The DFT of an imaginary signal is *conjugate antisymmetric*.

Properties of the DFT contd...

🟢 **Case V:** If $x(n)$ is *imaginary & circularly even*, that is, if

$$x(n) = -x^*(n) = x(\langle -n \rangle_N)$$

then, we obtain

$$\text{DFT}[x(n)] = \text{DFT}[-x^*(n)] = \text{DFT}[x(\langle -n \rangle_N)]$$

$$X(k) = -X^*(\langle -k \rangle_N) = X(\langle -k \rangle_N) \quad (\text{imaginary \& circularly even})$$

$$X(k) = -X^*(N - k) = X(N - k)$$

If $x(n)$ is imaginary & circularly even then its DFT $X(k)$ is also imaginary & circularly even.

Properties of the DFT contd...

🟢 **Case VI:** If $x(n)$ is *imaginary & circularly odd*, that is, if

$$x(n) = -x^*(n) = -x(\langle -n \rangle_N)$$

then, we obtain

$$\text{DFT}[x(n)] = \text{DFT}[-x^*(n)] = \text{DFT}[-x(\langle -n \rangle_N)]$$

$$X(k) = -X^*(\langle -k \rangle_N) = -X(\langle -k \rangle_N) \quad (\text{real \& circularly odd})$$

$$X(k) = -X^*(N - k) = -X(N - k)$$

If $x(n)$ is imaginary & circularly odd then its DFT $X(k)$ is real & circularly odd.

Properties of the DFT contd...

- **Case VII:** Circularly even and odd decomposition of real signals: Let $x(n)$ be a real signal and

$$x(n) \xleftrightarrow[N]{\text{DFT}} X(k)$$

- The DFT of the circularly even part of $x(n)$ is the real value of $X(k)$, that is,

$$x_{ce}(n) \xleftrightarrow[N]{\text{DFT}} X_R(k) = \text{Re}\{X(k)\}$$

- The DFT of the circularly odd part of $x(n)$ is the $j\text{Im}\{X(k)\}$, that is,

$$x_{co}(n) \xleftrightarrow[N]{\text{DFT}} jX_I(k) = j\text{Im}\{X(k)\}$$

Properties of the DFT contd...

● Duality:

$$\text{If } x(n) \xleftrightarrow[N]{\text{DFT}} X(k) \quad \text{then} \quad X(n) \xleftrightarrow[N]{\text{DFT}} N x(\langle -k \rangle_N)$$

● Proof:

$$\text{By definition} \quad x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j \frac{2\pi}{N} kn}$$

$$Nx(n) = \sum_{k=0}^{N-1} X(k) e^{j \frac{2\pi}{N} kn}$$

$$\text{Interchanging the variables } n \text{ and } k \quad Nx(k) = \sum_{n=0}^{N-1} X(n) e^{j \frac{2\pi}{N} kn}$$

$$\text{replacing } k \text{ by } -k \quad Nx(\langle -k \rangle_N) = \sum_{n=0}^{N-1} X(n) e^{-j \frac{2\pi}{N} kn} = \text{DFT}[X(n)]$$

Properties of the DFT contd...

● Circular Convolution (Multiplication of Two DFTs): If

$$x_1(n) \xleftrightarrow[N]{\text{DFT}} X_1(k) \quad \text{and} \quad x_2(n) \xleftrightarrow[N]{\text{DFT}} X_2(k)$$

$$\text{then} \quad x_3(n) = x_1(n) \circledast x_2(n) \xleftrightarrow[N]{\text{DFT}} X_3(k) = X_1(k)X_2(k)$$

where $\circledast$ denotes the N -point circular convolution. The circular convolution is defined as

$$x_3(n) = x_1(n) \circledast x_2(n) = \sum_{m=0}^{N-1} x_1(m)x_2(\langle n - m \rangle_N), \quad 0 \leq n \leq N - 1$$

● **Proof:**

$$\text{By definition} \quad X_3(k) = \sum_{n=0}^{N-1} x_3(n)W_N^{kn} = \sum_{n=0}^{N-1} [x_1(n) \circledast x_2(n)]W_N^{kn}$$

Properties of the DFT contd...

$$\begin{aligned}X_3(k) &= \sum_{n=0}^{N-1} \left(\sum_{m=0}^{N-1} x_1(m) x_2(\langle n - m \rangle_N) \right) W_N^{kn} \\&= \sum_{m=0}^{N-1} x_1(m) \left(\sum_{n=0}^{N-1} x_2(\langle n - m \rangle_N) W_N^{kn} \right) \\&= \sum_{m=0}^{N-1} x_1(m) \left(X_2(k) W_N^{km} \right) \\&= X_2(k) \sum_{m=0}^{N-1} x_1(m) W_N^{km} \\X_3(k) &= X_1(k) X_2(k)\end{aligned}$$

The multiplication of two DFTs of two sequences is equivalent to the circular convolution of the two sequences in the time domain.

Properties of the DFT contd...

Example: Perform the 4-point circular convolution of the following two sequences using analytical and graphical method.

$$x_1(n) = \{\underset{\uparrow}{1}, 2, 2\} \quad x_2(n) = \{\underset{\uparrow}{1}, 2, 3, 4\}$$

Solution: 1. Analytical Method:

$$x_3(n) = \sum_{m=0}^{N-1} x_1(m)x_2(\langle n-m \rangle_N) = \sum_{m=0}^3 x_1(m)x_2(\langle n-m \rangle_4), \quad 0 \leq n \leq 3$$

$$\begin{aligned} \text{for } n = 0: \quad x_3(0) &= \sum_{m=0}^3 x_1(m)x_2(\langle -m \rangle_4) \\ &= x_1(0)x_2(0) + x_1(1)x_2(\langle -1 \rangle_4) + x_1(2)x_2(\langle -2 \rangle_4) + x_1(3)x_2(\langle -3 \rangle_4) \\ &= x_1(0)x_2(0) + x_1(1)x_2(-1+4) + x_1(2)x_2(-2+4) + x_1(3)x_2(-3+4) \\ &= x_1(0)x_2(0) + x_1(1)x_2(3) + x_1(2)x_2(2) + x_1(3)x_2(1) \\ x_3(0) &= (1 \times 1) + (2 \times 4) + (2 \times 3) + (0 \times 2) = 1 + 8 + 6 + 0 = 15 \end{aligned}$$

Properties of the DFT contd...

$$\begin{aligned}\text{for } n = 1: \quad x_3(1) &= \sum_{m=0}^3 x_1(m)x_2(\langle 1 - m \rangle_4) \\ &= x_1(0)x_2(1) + x_1(1)x_2(0) + x_1(2)x_2(\langle -1 \rangle_4) + x_1(3)x_2(\langle -2 \rangle_4) \\ &= x_1(0)x_2(1) + x_1(1)x_2(0) + x_1(2)x_2(3) + x_1(3)x_2(2) \\ x_3(1) &= (1 \times 2) + (2 \times 1) + (2 \times 4) + (0 \times 3) = 2 + 2 + 8 + 0 = 12\end{aligned}$$

$$\begin{aligned}\text{for } n = 2: \quad x_3(2) &= \sum_{m=0}^3 x_1(m)x_2(\langle 2 - m \rangle_4) \\ &= x_1(0)x_2(2) + x_1(1)x_2(1) + x_1(2)x_2(0) + x_1(3)x_2(\langle -1 \rangle_4) \\ &= x_1(0)x_2(2) + x_1(1)x_2(1) + x_1(2)x_2(0) + x_1(3)x_2(3) \\ x_3(2) &= (1 \times 3) + (2 \times 2) + (2 \times 1) + (0 \times 4) = 3 + 4 + 2 + 0 = 9\end{aligned}$$

$$\begin{aligned}\text{for } n = 3: \quad x_3(3) &= \sum_{m=0}^3 x_1(m)x_2(\langle 3 - m \rangle_4) \\ &= x_1(0)x_2(3) + x_1(1)x_2(2) + x_1(2)x_2(1) + x_1(3)x_2(0) \\ x_3(3) &= (1 \times 4) + (2 \times 3) + (2 \times 2) + (0 \times 1) = 4 + 6 + 4 + 0 = 14\end{aligned}$$

Therefore, $x_3(n) = x_1(n) \circledast x_2(n) = \{15, 12, 9, 14\}$.
Digital Signal Processing

Properties of the DFT contd...

Therefore, $x_3(n) = x_1(n) \circledast x_2(n) = \{\underset{\uparrow}{15}, 12, 9, 14\}$.

2. Graphical Method: Change the variables of the two sequences from n to m to get $x_1(m) = \{\underset{\uparrow}{1}, 2, 2, 0\}$ and $x_2(m) = \{\underset{\uparrow}{1}, 2, 3, 4\}$. Graph each sequence as points on a circle [Fig. (a)].

$$x_3(n) = \sum_{m=0}^3 x_1(m)x_2(\langle n - m \rangle_4), \quad 0 \leq n \leq 3$$

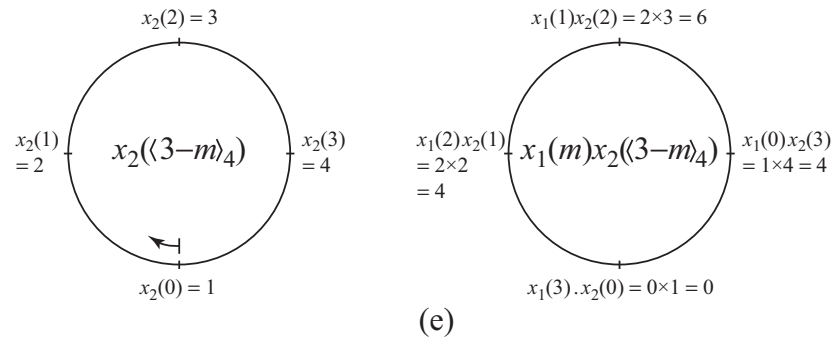
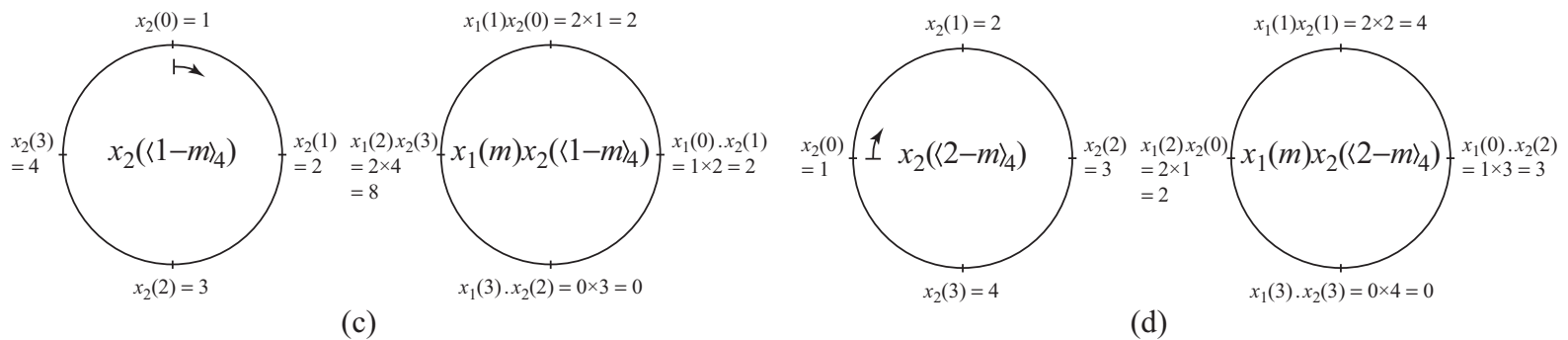
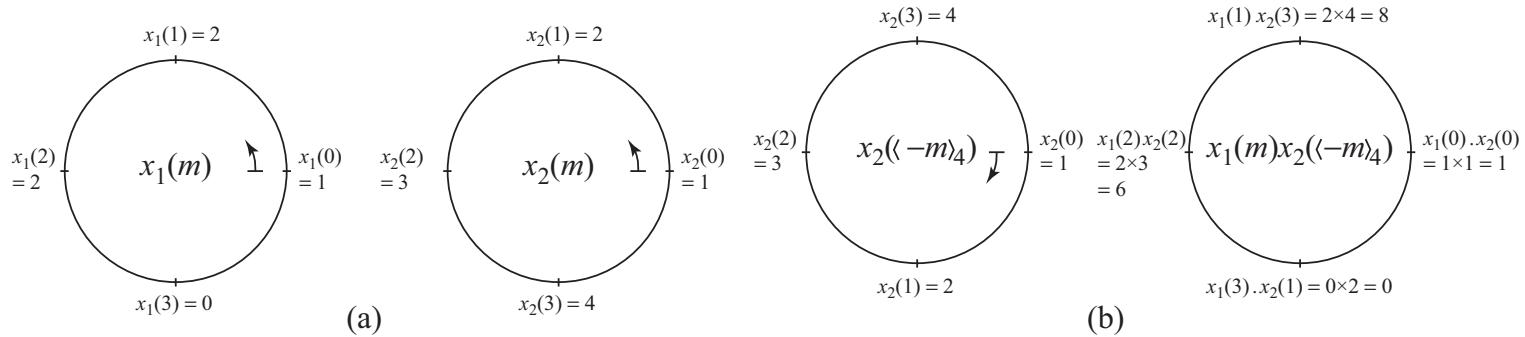
For $n = 0$, we have

$$x_3(0) = \sum_{m=0}^3 x_1(m)x_2(\langle -m \rangle_4)$$

The folded sequence $x_2(\langle -m \rangle_4)$ and the product sequence $x_1(m)x_2(\langle -m \rangle_4)$ are shown in Fig. (b). Finally

$$x_3(0) = 1 + 8 + 6 + 0 = 15$$

Properties of the DFT contd...



Properties of the DFT contd...

For $n = 1$, we have

$$x_3(1) = \sum_{m=0}^3 x_1(m)x_2(\langle 1 - m \rangle_4)$$

The sequence $x_2(\langle 1 - m \rangle_4)$ is simply the sequence $x_2(\langle -m \rangle_4)$ rotated anticlockwise by one unit in time as illustrated in Fig. (c). The product sequence $x_1(m)x_2(\langle 1 - m \rangle_4)$ is also illustrated in Fig. (c). Thus $x_3(1) = 2 + 2 + 8 + 0 = 12$.

For $n = 2$, we have

$$x_3(2) = \sum_{m=0}^3 x_1(m)x_2(\langle 2 - m \rangle_4)$$

The sequence $x_2(\langle 2 - m \rangle_4)$ is simply the sequence $x_2(\langle -m \rangle_4)$ rotated anticlockwise by two units in time as illustrated in Fig. (d). The product sequence $x_1(m)x_2(\langle 2 - m \rangle_4)$ is also illustrated in Fig. (d).

Properties of the DFT contd...

Thus $x_3(2) = 3 + 4 + 2 + 0 = 9$.

For $n = 3$, we have

$$x_3(3) = \sum_{m=0}^3 x_1(m)x_2(\langle 3 - m \rangle_4)$$

The sequence $x_2(\langle 3 - m \rangle_4)$ is simply the sequence $x_2(\langle -m \rangle_4)$ rotated anticlockwise by three units in time as illustrated in Fig. (e). The product sequence $x_1(m)x_2(\langle 3 - m \rangle_4)$ is also illustrated in Fig. (e). Thus

$$x_3(3) = 4 + 6 + 4 + 0 = 14$$

Therefore, $x_3(n) = \{15, 12, 9, 14\}$.

Properties of the DFT contd...

● **Circular Correlation:** In general, for complex-valued sequences $x(n)$ and $y(n)$, if

$$x(n) \xleftrightarrow[N]{\text{DFT}} X(k) \quad \text{and} \quad y(n) \xleftrightarrow[N]{\text{DFT}} Y(k)$$

$$\text{then} \quad r_{xy}(l) \xleftrightarrow[N]{\text{DFT}} R_{xy}(k) = X(k)Y^*(k)$$

● **Proof:** $r_{xy}(l)$ is the circular convolution of $x(n)$ and $y^*(\langle -n \rangle_N)$, that is

$$r_{xy}(l) = x(n) \circledast y^*(\langle -n \rangle_N) = \sum_{n=0}^{N-1} x(n)y^*(\langle n-l \rangle_N)$$

$$\text{Taking the DFT, we get} \quad R_{xy}(k) = X(k)Y^*(k)$$

$$\text{If } y(n) = x(n) \quad \text{then} \quad r_{xx}(l) \xleftrightarrow[N]{\text{DFT}} R_{xx}(k) = X(k)X^*(k) = |X(k)|^2$$

Properties of the DFT contd...

● Multiplication (or Modulation) Property:

$$\text{If } x_1(n) \xleftrightarrow[N]{\text{DFT}} X_1(k) \quad \text{and} \quad x_2(n) \xleftrightarrow[N]{\text{DFT}} X_2(k)$$
$$\text{then } x_1(n)x_2(n) \xleftrightarrow[N]{\text{DFT}} \frac{1}{N}[X_1(k) \circledast X_2(k)]$$

● **Proof:** By definition

$$\begin{aligned} \text{DFT}[x_1(n)x_2(n)] &= \sum_{n=0}^{N-1} x_1(n)x_2(n)W_N^{kn} = \sum_{n=0}^{N-1} \left(\frac{1}{N} \sum_{m=0}^{N-1} X_1(m)W_N^{-mn} \right) x_2(n)W_N^{kn} \\ &= \frac{1}{N} \sum_{m=0}^{N-1} X_1(m) \left(\sum_{n=0}^{N-1} x_2(n)W_N^{(k-m)n} \right) \\ &= \frac{1}{N} \sum_{m=0}^{N-1} X_1(m)X_2(\langle k-m \rangle_N) \\ \text{DFT}[x_1(n)x_2(n)] &= \frac{1}{N}[X_1(k) \circledast X_2(k)] \end{aligned}$$

Properties of the DFT contd...

● **Parseval's Relation:** For complex-valued sequences $x(n)$ and $y(n)$, in general, if

$$x(n) \xleftrightarrow[N]{\text{DFT}} X(k) \quad \text{and} \quad y(n) \xleftrightarrow[N]{\text{DFT}} Y(k)$$

then

$$\sum_{n=0}^{N-1} x(n)y^*(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k)Y^*(k)$$

● **Proof:** Consider the LHS of the above equation, we have

$$\begin{aligned} \sum_{n=0}^{N-1} x(n)y^*(n) &= \sum_{n=0}^{N-1} x(n)[y(n)]^* \\ &= \sum_{n=0}^{N-1} x(n) \left[\frac{1}{N} \sum_{k=0}^{N-1} Y(k)e^{j\frac{2\pi}{N}kn} \right]^* \end{aligned}$$

Properties of the DFT contd...

$$\begin{aligned}\sum_{n=0}^{N-1} x(n)y^*(n) &= \sum_{n=0}^{N-1} x(n) \left[\frac{1}{N} \sum_{k=0}^{N-1} Y^*(k) e^{-j\frac{2\pi}{N}kn} \right] \\ &= \frac{1}{N} \sum_{k=0}^{N-1} Y^*(k) \left[\sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi}{N}kn} \right] \\ &= \frac{1}{N} \sum_{k=0}^{N-1} X(k) Y^*(k)\end{aligned}$$

If $y(n) = x(n)$, the above equation reduces to

$$\sum_{n=0}^{N-1} |x(n)|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X(k)|^2$$

Filtering of Long Data Sequences Using DFT

- Two Methods that are used for filtering of long data sequences using DFT are the *overlap-save method* and the *overlap-add method*.

Filtering of Long Data Sequences Using DFT

- Two Methods that are used for filtering of long data sequences using DFT are the *overlap-save method* and the *overlap-add method*.

- Overlap-Save Method:**

- Let $h(n)$ be the impulse response of length M . (Let $M = 3$)
- Let the length of $x(n)$ be 10 which is greater than M .
- We can divide $x(n)$ into sections (or blocks) $x_r(n)$ of length- N . (Let $N = 6$)
- Each input section overlaps the preceding section by $(M - 1)$ samples. To begin the processing, the first $(M - 1)$ samples of the first section are set to zero. Therefore, the input sections are

$$x_0(n) = \{0, 0, x(0), x(1), x(2), x(3)\}$$

$$x_1(n) = \{x(2), x(3), x(4), x(5), x(6), x(7)\}$$

$$x_2(n) = \{x(6), x(7), x(8), x(9), 0, 0\}$$

Overlap-Save Method contd...

- The circular convolution of each section $x_r(n)$ with $h(n)$ is denoted by $y_r(n) = x_r(n) \circledast h(n)$.

$$y_0(n) = x_0(n) \circledast h(n) = \underbrace{\{y_0(0), y_0(1), y_0(2), y_0(3), y_0(4), y_0(5)\}}_{\text{discard}}$$

$$y_1(n) = x_1(n) \circledast h(n) = \underbrace{\{y_1(0), y_1(1), y_1(2), y_1(3), y_1(4), y_1(5)\}}_{\text{discard}}$$

$$y_2(n) = x_2(n) \circledast h(n) = \underbrace{\{y_2(0), y_2(1), y_2(2), y_2(3), y_2(4), y_2(5)\}}_{\text{discard}}$$

- The first $(M - 1)$ samples of each output section must be discarded and the remaining $(N - M + 1)$ samples constitute the desired result from linear convolution. The remaining $(N - M + 1)$ samples from each successive sections are then concatenated to reconstruct the final filtered output.

$$y(n) = \{y_0(2), y_0(3), y_0(4), y_0(5), y_1(2), y_1(3), y_1(4), y_1(5), y_2(2), y_2(3), y_2(4), y_2(5)\}$$

Overlap-Save Method contd...

- The reason for the name *overlap-save* method is that the input $x(n)$ is divided into overlapping sections of length- N and after performing the N -point circular convolution, only the last $(N - M + 1)$ samples are saved and concatenated to determine the linear convolution result.

Example: The impulse response of LTI system is $h(n) = \{\underset{\uparrow}{1}, 0, -1\}$. Find the response of the system if the input is $x(n) = (n + 1), 0 \leq n \leq 9$. Use overlap-save method.

Solution: Since $h(n)$ has length $M = 3$, we will have to overlap each section with the previous one by $(M - 1) = 2$ samples. Let the length of each section is $N = 6$. The first $(M - 1) = 2$ samples of the first section are set to zero. Thus the sections (or blocks) of input sequence are

$$x_0(n) = \{0, 0, 1, 2, 3, 4\}$$

$$x_1(n) = \{3, 4, 5, 6, 7, 8\}$$

$$x_2(n) = \{7, 8, 9, 10, 0, 0\}$$

Overlap-Save Method contd...

Now we will compute the 6-point circular convolution of each section $x_r(n)$ with $h(n)$ to get $y_r(n) = x_r(n) \circledast h(n)$.

$$y_0(n) = x_0(n) \circledast h(n) = \{\underbrace{-3, -4}_{\text{discard}}, 1, 2, 2, 2\}$$

$$y_1(n) = x_1(n) \circledast h(n) = \{\underbrace{-4, -4}_{\text{discard}}, 2, 2, 2, 2\}$$

$$y_2(n) = x_2(n) \circledast h(n) = \{\underbrace{7, 8}_{\text{discard}}, 2, 2, -9, -10\}$$

The first $M - 1 = 2$ samples of each output section $y_r(n)$ must be discarded. The remaining $N - M + 1 = 4$ samples from each successive sections $y_r(n)$ are then concatenated to reconstruct the final filtered output $y(n)$.

$$y(n) = \{\underset{\uparrow}{1}, 2, 2, 2, 2, 2, 2, 2, 2, -9, -10\}$$

Overlap-Add Method

● Overlap-Add Method:

- Let $h(n)$ be the impulse response of length M . (Let $M = 3$)
- Let the length of $x(n)$ be 10 which is greater than M .
- We can divide $x(n)$ into sections (or blocks) $x_r(n)$ of length- N . (Let $N = 4$)
- Since $h(n)$ is of length $M = 3$ and $x_r(n)$ is of length $N = 4$, the linear convolution is of length $(N + M - 1) = (4 + 3 - 1) = 6$. We have to pad each section by $(M - 1) = 2$ zeros and $h(n)$ by $(N - 1) = 3$ zeros so that their lengths will be equal to the length of the result of the linear convolution. Therefore, the input sections are

$$x_0(n) = \{x(0), x(1), x(2), x(3), \underbrace{0, 0}_{2 \text{ zeros}}\}$$

$$x_1(n) = \{x(4), x(5), x(6), x(7), \underbrace{0, 0}_{2 \text{ zeros}}\}$$

$$x_2(n) = \{x(8), x(9), 0, 0, \underbrace{0, 0}_{2 \text{ zeros}}\}$$

Overlap-Add Method contd...

- Now we will compute the linear convolution using 6-point circular convolution of each section $x_r(n)$ with $h(n)$ to get $y_r(n)$.

$$y_0(n) = h(n) \circledast x_0(n) = \{y_0(0), y_0(1), y_0(2), y_0(3), \underbrace{y_0(4), y_0(5)}_{\text{added}}\}$$

$$y_1(n) = h(n) \circledast x_1(n) = \{\underbrace{y_1(0), y_1(1)}_{\text{added}}, y_1(2), y_1(3), \underbrace{y_1(4), y_1(5)}_{\text{added}}\}$$

$$y_2(n) = h(n) \circledast x_2(n) = \{\underbrace{y_2(0), y_2(1)}_{\text{added}}, y_2(2), y_2(3), y_2(4), y_2(5)\}$$

- The filtered output sections will overlap by $(M - 1) = 2$ samples and these overlap samples must be added to get the correct final result.

$$y(n) = \{y_0(0), y_0(1), y_0(2), y_0(3), y_0(4) + y_1(0), y_0(5) + y_1(1), y_1(2), y_1(3), \\ y_1(4) + y_2(0), y_1(5) + y_2(1), y_2(2), y_2(3), y_2(4), y_2(5)\}$$

Overlap-Add Method contd...

- This procedure is called the *overlap-add method* because the output filtered sections are overlapped and added to get the correct final result.

Example: The impulse response of LTI system is $h(n) = \{1, 0, -1\}$. Find the response of the system if the input is $x(n) = (n + 1), 0 \leq n \leq 9$. Use overlap-save method.

Solution: The total length of the input sequence $x(n)$ is 10 and the length of the impulse response $h(n)$ is $M = 3$. Let the length of each section $x_r(n)$ is $N = 4$. Since $h(n)$ is of length $M = 3$ and $x_r(n)$ is of length $N = 4$, the linear convolution is of length $(N + M - 1) = (4 + 3 - 1) = 6$. We have to pad each section by $(M - 1) = 2$ zeros and $h(n)$ by $(N - 1) = 3$ zeros so that their lengths will be equal to the length of the result of the linear convolution. Thus the sections (or blocks) of input sequence are

$$x_0(n) = \{1, 2, 3, 4, \underbrace{0, 0}_{2 \text{ zeros}}\}$$

$$x_1(n) = \{5, 6, 7, 8, \underbrace{0, 0}_{2 \text{ zeros}}\}$$

$$x_2(n) = \{9, 10, 0, 0, \underbrace{0, 0}_{2 \text{ zeros}}\}$$

Overlap-Add Method contd...

Now we will compute the linear convolution using 6-point circular convolution of each section $x_r(n)$ with $h(n)$ to get $y_r(n)$.

$$y_0(n) = h(n) \circledast x_0(n) = \{1, 2, 2, 2, \underbrace{-3, -4}_{\text{added}}\}$$

$$y_1(n) = h(n) \circledast x_1(n) = \{\overbrace{5, 6}^{\text{added}}, 2, 2, \underbrace{-7, -8}_{\text{added}}\}$$

$$y_2(n) = h(n) \circledast x_2(n) = \{\overbrace{9, 10}^{\text{added}}, -9, -10, 0, 0\}$$

The filtered output sections will overlap by $(M - 1) = 2$ samples and these overlap samples must be added to get the correct final result.

$$\begin{aligned} y(n) &= \{\underset{\uparrow}{1}, 2, 2, 2, -3 + 5, -4 + 6, 2, 2, -7 + 9, -8 + 10, -9, -10\} \\ &= \{\underset{\uparrow}{1}, 2, 2, 2, 2, 2, 2, 2, 2, 2, -9, -10\} \end{aligned}$$

Discrete Fourier Transform

Thank You!